

experiments (Fig. 3A), the contralateral and antipodal strategies form straight lines because the predicted $\theta_i^{(n)}$ are uniquely determined by the predator angular position ϕ_i , while the other distributions are spread out due to their dependency on the predator heading ψ_i and λ_i .

To measure the difference between model predictions and experimental data, we estimated numerically the Kullback–Leibler (K–L) divergence $\mathcal{D}_{\text{KL}}$, which quantifies the entropy of $p^{(n)}(\mathbf{s}, \mathbf{r})$ relative to $p_o(\mathbf{s}, \mathbf{r})$, using the method in (33); *SI Appendix, S5*. Results of the K–L divergence are shown in Fig. 4A for all five strategies applied to the slow, mid-speed, and fast predator, as well as the combined data. The actual K–L divergence is always nonnegative; the negative values are due to the discrete estimation of the PDF. In each of the four datasets, the distance-optimal and orthogonal strategies yield the lowest estimates of the K–L divergence, implying that, of all five evasion strategies, they give the closest predictions of the prey escape response. The distance-optimal strategy performs slightly better for the slow and mid-speed predator, while the orthogonal is more advantageous for the fast predator and when considering all data combined. The antipodal strategy also gives relatively low K–L divergence estimates. The parallel and contralateral strategies, whose K–L divergence estimates are significantly higher than the other strategies, have the worst fit to experimental data across all predator speeds.

Modeling noise in sensing and response. We next introduced sensing and response noise according to Eq. 5. To model sensing noise, we considered $\hat{\mathbf{s}}$ to be normally distributed around the actual state of the predator $\mathbf{s}$, with dispersion $\sigma_{\mathbf{S}}$, and to model response noise, we considered $\mathbf{r}$ to be normally distributed around the desired response $\hat{\mathbf{r}}$, with dispersion $\sigma_{\mathbf{R}}$. Substituting the noise models $p(\hat{\mathbf{s}}|\mathbf{s}; \sigma_{\mathbf{S}})$ and $p(\mathbf{r}|\hat{\mathbf{r}}; \sigma_{\mathbf{R}})$ into Eq. 5, and recalling that $p^{(n)}(\hat{\mathbf{r}}|\hat{\mathbf{s}}) = \delta(\hat{\mathbf{r}} - \mathbf{f}^{(n)}(\hat{\mathbf{s}}))$, we arrived, for each evasion strategy n , at a probabilistic model that depends on the noise parameters $\sigma = \{\sigma_{\mathbf{S}}, \sigma_{\mathbf{R}}\}$ (*SI Appendix, section 4*). Specifically, we used a von Mises distribution (normal distribution on the circle) for θ , ϕ , and λ with noise parameters σ_{Θ} , σ_{Φ} , and σ_{Λ} ; we let the noise on ψ follow from $\psi = \phi + \lambda + \pi$ (*SI Appendix, section 4*). At zero noise, the von Mises distribution converges to a Dirac-delta function at the mean value; when the noise level is high,

it approaches a circular uniform distribution with constant PDF $1/(2\pi)$ in any escape direction.

Limit of high noise levels. If the response noise σ_{Θ} is large, any evasion direction is predicted with equal probability density $1/(2\pi)$, irrespective of the strategy or the sensing noise, that is, all strategies become essentially equivalent to the pure-protean strategy. On the other hand, if the response is precise $\sigma_{\Theta} = 0$, but the noise in sensing the predator's angular position σ_{Φ} is large, all strategies, except the contralateral, converge to the pure-protean strategy; the contralateral strategy predicts $\theta = \pm\pi/2$ with equal probability. If the prey's response and sensing of the predator angular position are both precise $\sigma_{\Theta} = \sigma_{\Phi} = 0$, but the prey's sensing of the predator's heading is very noisy (σ_{Λ} large), the antipodal and contralateral strategies do not get affected while the parallel strategy becomes protean. Interestingly, in this case, the distance-optimal strategy predicts higher probability of evasion in directions opposite to the predator location spanning a range of 2χ (*SI Appendix, section 4 and Fig. S9*). That is, the distance-optimal strategy becomes a noisy variant of the antipodal strategy. For $\chi = \pi/2$, the orthogonal strategy with large noise on λ converges to the antipodal strategy with uniform noise spanning a range of π on either ϕ or θ .

Optimizing noise levels in sensing and response. For each noisy evasion strategy, we calculated the noise parameters $\sigma = \{\sigma_{\mathbf{S}}, \sigma_{\mathbf{R}}\}$ that maximize the total likelihood $\mathcal{L}$ of the model given an experimental dataset, or equivalently minimize the negative log-likelihood function NLL (*SI Appendix, section 6*)

$$\text{NLL} = -\ln \mathcal{L}(\sigma | \{\mathbf{r}|\mathbf{s}\}; n) = -\sum_i \ln p^{(n)}(\mathbf{r}_i | \mathbf{s}_i; \sigma), \quad [7]$$

where $p^{(n)}(\mathbf{r}_i | \mathbf{s}_i; \sigma)$ is the conditional PDF of obtaining a response $\mathbf{r}_i$ given stimulus $\mathbf{s}_i$ for strategy n at noise level σ . The optimal noise parameters σ^* are given by

$$\sigma^* = \arg \min_{\sigma} \text{NLL}. \quad [8]$$

We solved this optimization problem numerically in the range $\sigma_{\Phi}, \sigma_{\Lambda}, \sigma_{\Theta} \in (0, \pi)$ (*SI Appendix, Fig. S6*). In Fig. 3D, we plot realizations generated from the five probabilistic evasion models $p^{(n)}(\mathbf{s}, \mathbf{r}; \sigma^*)$ at the optimal noise values corresponding

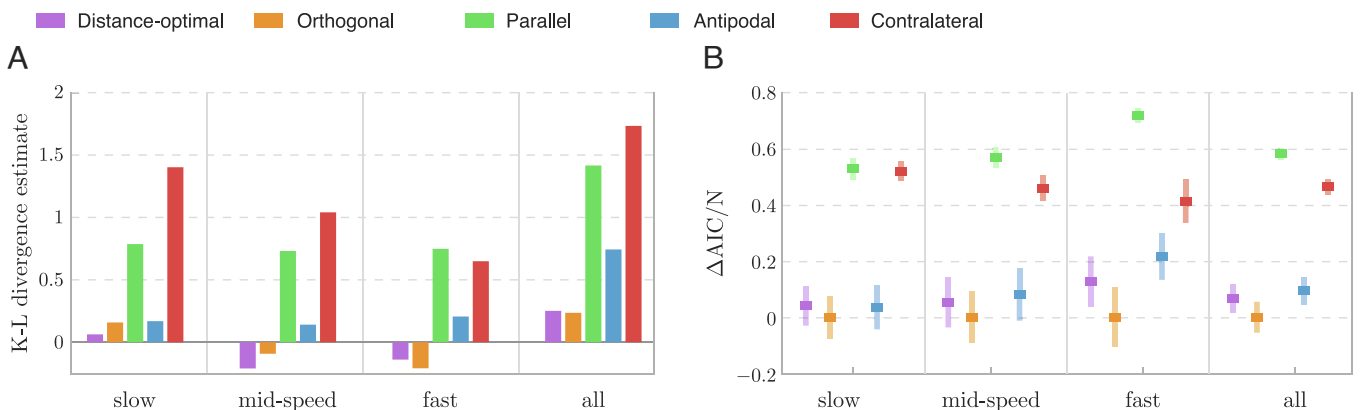


Fig. 4. Evaluation of precise and noisy evasion strategies. (A) K–L divergence estimate from precise model predictions to experiment data is computed separately for each dataset (slow, mid-speed, and fast predator) and for all data combined. The K–L values for the distance-optimal and orthogonal strategies are the lowest, indicating better fit to data. (B) AIC difference ($\Delta\text{AIC} = \text{AIC} - \text{AIC}_{\min}$), normalized by the respective sample size of each dataset. For each dataset, we used the bootstrap method to construct 200 distinct datasets (by sampling with repetition) of equal size to the original dataset. We optimized each of 200 sets, evaluated the corresponding AIC, and computed the mean and standard deviation of ΔAIC . The orthogonal strategy has the lowest ΔAIC , indicating that it is the most parsimonious strategy and best explains the data.

to the dataset of all predator speeds combined. Compared to the deterministic predictions in Fig. 3B, all five distributions appear closer to the experimental data in Fig. 3A.

Evaluating strategies under optimal noise parameters. To evaluate how well each optimized strategy describes the experimental data, we applied the Akaike information criterion (AIC) defined as ref. 35

$$\text{AIC} = 2K - 2 \ln \mathcal{L}(\sigma^* | (\mathbf{s}, \mathbf{r}); n), \quad [9]$$

where K is the number of model parameters in each strategy. AIC considers both the goodness of fit represented by the likelihood function and the complexity of the model: If two models have the same likelihood to explain the data, the criterion favors the simpler model. For example, for the antipodal strategy, we have two noise parameters $\sigma_{\mathbf{S}} \equiv \{\sigma_{\Phi}\}$ and $\sigma_{\mathbf{R}} \equiv \{\sigma_{\Theta}\}$, thus $K = 2$; in contrast, for the orthogonal strategy, we have three noise parameters, $\sigma_{\mathbf{S}} \equiv \{\sigma_{\Phi}, \sigma_{\Lambda}\}$ and $\sigma_{\mathbf{R}} \equiv \{\sigma_{\Theta}\}$, and the orthogonal strategy is deemed more complex than the antipodal strategy.

We used bootstrapping to probe the accuracy of our evaluation of the noisy strategies. Starting from each dataset (e.g., that of the fast predator), we constructed 200 distinct datasets of equal size to the original dataset (e.g., N_{fast}) by random sampling with repetition. We solved the optimization problem 200 times and obtained 200 values of σ^* per strategy for each dataset. We calculated the likelihood value $\mathcal{L}$ and evaluated the AIC for all 200 optimal noise values, thus obtaining a distribution of AIC values for each strategy and predator speed. The mean and standard deviation of the distributions of AIC values, minus the lowest mean value, and normalized by the size of the respective dataset (Fig. 4B) show that strategies with lower mean values of the AIC better fit the experimental data.

The results based on the AIC evaluation of the probabilistic strategies in the presence of sensory and response noise are mostly consistent with the results based on the K–L divergence (Fig. 4A) for precise sensing and response, but with marked differences. The orthogonal strategy ranks the highest in every dataset; the distance-optimal strategy is slightly behind, in second place, in all but the slow predator dataset where the antipodal strategy ranks second. The difference between the orthogonal, distance-optimal, and antipodal strategies is most distinguishable in the case of the fast predator. The contralateral and parallel strategies come last in all datasets and are least descriptive of experimental data.

Further analysis of distance-optimal strategy. While the predator speed was controlled at $V = 2, 11, 20 \text{ cm s}^{-1}$, the zebrafish larvae were almost identical in all experiments, implying that the speed ratio U/V varied drastically between evasion instances: For the fast predator, this ratio is up to 10 times that of the slow predator. If the prey were to sense and use the speed ratio to implement the distance-optimal strategy, we would expect the best performance to appear at different values of $\chi = \cos^{-1}(U/V)$, depending on predator speed. To test this, we evaluated this strategy for the slow, mid, and fast predator as a function of $\chi \in [0, 90^\circ]$ under both precise and noisy sensing and response (SI Appendix, section 7 and Fig. S14). We found that the K–L divergence decreased as χ increased and reached a minimum near $\chi = 75^\circ$ independent of predator speed. Similarly, the NLL dropped as χ increased until it reached a minimum at, or close to, $\chi = 90^\circ$. These results suggest that, even if following the distance-optimal strategy, the prey does not rely on real-time and accurate measurements of the speed ratio U/V but favors the limit of large predator speed ($\chi \rightarrow 90^\circ$), where the distance-optimal strategy converges to the orthogonal strategy.

The same conclusion can be reached by examining the values of the optimized noise parameters. In the range $20^\circ \lesssim \chi \lesssim 75^\circ$, the optimizer mostly selects the largest possible value of $\sigma_{\Lambda} = \pi$ to best fit the data. This high level of optimized noise indicates that λ is not an effective sensory cue in the distance-optimal strategy and that the prey is unlikely to use this strategy at moderate χ values (SI Appendix, section 7 and Fig. S14).

Evaluating the Biomechanical Constraints on Escape Strategy.

To complete our evaluation of fish evasion strategies, we considered the biomechanics of the C-start response. In ref. 34, the motion of a zebrafish larvae undergoing a C-start maneuver starting from a straight motionless configuration was recorded using high-speed photography, and the time evolution of each segment of the fish body from the onset of evasion at time $t = 0$ to after the completion of the C-start response at $t = T = 25 \text{ ms}$ was measured. We developed a mathematical model of the biomechanics of these events and incorporated that model into our analysis.

We reinterpreted the experimental measurements in the context of a three-link fish, head, middle, and tail (Fig. 5B), and we extracted from experimental measurements the fish orientation $\beta(t)$ and rotations $\alpha_1(t)$ and $\alpha_2(t)$ of the head and tail relative to the middle segment (SI Appendix, sections 8–9). The time evolution of the zebrafish body during evasion follows the three archetypal stages of the C-start response: In stage 1, the fish curls its body to one side, rapidly unfurls its body in stage 2, and begins its undulatory swimming in stage 3 (Fig. 5C and D).

A larger number of C-start maneuvers were recorded in ref. 23, albeit only measuring the maximum degree of body bending α_{max} and the net change in heading $\theta = \beta(T) - \beta(0)$ induced by the C-start maneuver (Fig. 5F). These results show that the change in heading direction θ correlates strongly with the degree of body bending (23). In all recorded maneuvers, the change in body orientation barely reaches 100° .

Physics-based modeling of the C-start response. To shed light on the relationship between body deformations and change in heading θ during evasion, we employed a physics-based model of a three-link fish in potential flow (36, 37). Experimental and computational flow analysis had shown that the C-start maneuver is dominated by unsteady, pressure-based exchange of momentum between the fish and surrounding fluid, with negligible contributions from fluid viscosity and shed vorticity (26, 38). The potential flow model captures these unsteady pressure forces via the added mass effect (SI Appendix, S8). The fish model is composed of three identical prolate spheroids (of major and minor axes a and b) such that the head and tail are free to rotate relative to the middle link (Fig. 5B); as before, body deformations are described by the angles $\alpha_1(t)$, $\alpha_2(t)$ representing the relative head and tail rotations as a function of time t , and body orientation $\beta(t)$ is the angle between the middle section and an inertial direction taken along the direction of the initially straight fish.

From consideration of momentum balance on the fish-fluid system, we arrived at an equation governing the rate of change of body orientation (37, 39, 40) (SI Appendix, section 8)

$$\dot{\beta} = A_1(\alpha_1, \alpha_2)\dot{\alpha}_1 + A_2(\alpha_1, \alpha_2)\dot{\alpha}_2, \quad [10]$$

where A_1 and A_2 are nonlinear functions of body deformations $\alpha_1(t)$, $\alpha_2(t)$; A_1 and A_2 also depend on fish geometry and fluid and body densities (ρ_f and ρ_b). For $\rho_b = \rho_f$, the fish is neutrally buoyant. When the fluid forces dominate, the fish can be considered massless, and ρ_b is set to zero. Body rotations

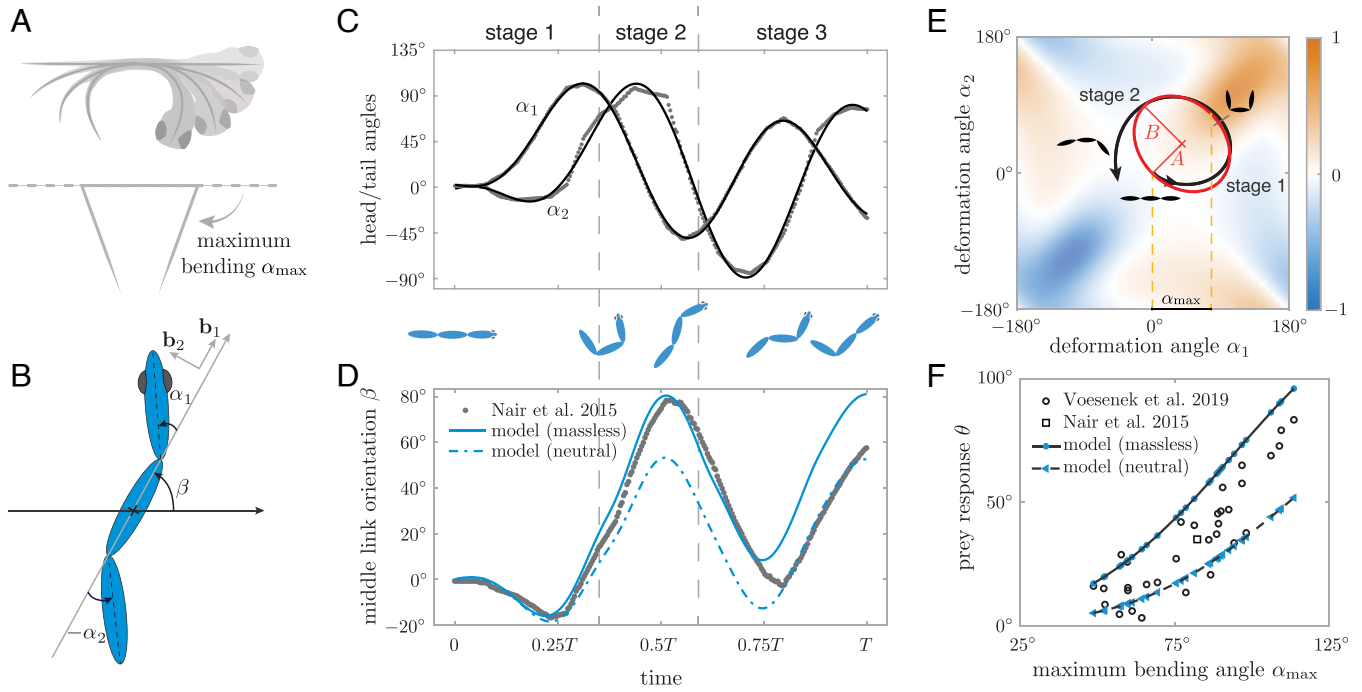


Fig. 5. Biomechanics of fish C-start response. (A) Larval zebrafish bends its body into a C-shape to initiate a fast start. (B) Three-fish model, with α_1 and α_2 representing the fish body shape and β the overall body orientation relative to the straight pre-evasion direction. (C) Experimental data of shape changes of larval zebrafish during evasion taken from ref. 34 and processed to represent body deformations in terms of head and tail rotations α_1, α_2 (gray dots) then fitted by third-order Fourier series (black lines). (D) Experimental data of overall body rotation for the same evasion instance shown in C (gray dots and black line). Predictions based on the fish model, taking as input the shape changes in C, are shown in blue lines (solid line for massless and dashed line for neutrally buoyant fish). (E) The sequence of shape changes in C forms a curve C in the shape space (α_1, α_2) (black line). The experimental curve C is approximated by an ellipse (red) of axes A and B along $\alpha_1 = \pm \alpha_2$ directions. Colormap represents $\text{curl}_2[A_1, A_2]$ of the fish model, which predicts larger turns for curves that encompass solely positive (orange) or negative (blue) values. (F) By varying A and calculating B that maximizes the turn, we get a mapping from maximum bending angle $\alpha_{\max} = \sqrt{2}A$ to turning angle θ for massless and neutrally buoyant fish (blue lines) that form upper and lower bounds on the experimental dataset of ref. 23. Both numerical and experimental data show that the C-start mechanics limits larval zebrafish to turning angles $\theta \lesssim 100^\circ$.

are proportional to the line integral of Eq. 10 over a curve C describing body deformations in the shape space (α_1, α_2) . For a closed curve C, this line integral can be rewritten, using the Stokes theorem, as an area integral over the region of the (α_1, α_2) space enclosed by C,

$$\theta = \beta(T) - \beta(0) = \iint \left(\frac{\partial A_2}{\partial \alpha_1} - \frac{\partial A_1}{\partial \alpha_2} \right) d\alpha_1 d\alpha_2. \quad [11]$$

The scalar field $\text{curl}_2([A_1, A_2]) \equiv \partial A_2 / \partial \alpha_1 - \partial A_1 / \partial \alpha_2$ is shown in Fig. 5E as a colormap over the entire shape space (α_1, α_2) . To maximize the turning angle θ , a straight fish should deform its body following a closed curve C that encompasses either nonpositive or nonnegative values of $\text{curl}_2([A_1, A_2])$, i.e., either blue or orange regions of the shape space. Closed curves in the orange region lead to turning counterclockwise. By symmetry, diagonally opposite curves in the blue region lead to turning clockwise. Theoretically, the simplest curve for turning is a circle or an ellipse in the shape space of major axis A aligned with $\alpha_1 = \alpha_2$, for which the maximum bending angle is $\alpha_{\max} = \sqrt{2}A$ (Fig. 5E). Corresponding fish shape deformations and body rotations $\beta(t)$ are discussed in SI Appendix, (S8-9 and Figs. S15-S16).

Comparing model predictions to C-start induced turning of the fish body. We represented the empirical time evolution of shape deformations $(\alpha_1(t), \alpha_2(t))$ (Fig. 5C) onto the shape space (Fig. 5E). Interestingly, the curve C (black line) traced by the actual fish follows closely the elliptic curve (red line) predicted by the model as best for turning. Moreover, when taking the

empirical values of $\alpha_1(t)$ and $\alpha_2(t)$ as input to the physics-based model in Eq. 10, the resulting predictions of $\beta(t)$ (blue lines in Fig. 5D) follow closely the empirical values of $\beta(t)$ (black line), especially during the first stage of the C-start response, where vorticity is negligible; note that while the neutrally buoyant model (dashed blue line) deviates slightly from empirical observations in stage 2, the massless fish model (solid blue line) performs remarkably well way into stage 3, indicating that indeed unsteady pressure forces dominate the C-start maneuver, as previously predicted (26).

We next considered a family of shape changes following the elliptic curve in Fig. 5E by varying A such that $\alpha_{\max} = \sqrt{2}A$ varied from 0 to 120° . This upper limit on $\alpha_{\max}$ corresponds to a maximum bending angle without causing the head and tail of the model fish to cross each other and is consistent with the experimental observations of ref. 23. Using Eq. 11, we computed, for each $\alpha_{\max}$, the value of B that optimizes the change in orientation θ , thus creating a map from $\alpha_{\max}$ to θ . We compare these model predictions (blue lines) to experimental data (23) (black dots) in Fig. 5F. As before, we considered massless and neutrally buoyant fish. The theoretical predictions behave nearly as upper and lower limits to experimental data. As observed previously (23), turning in the model fish barely reaches 100° even when the three-link fish bends its body to the extreme of the head and tail touching. This indicates that the biomechanics of the C-start maneuver imposes an upper limit on achievable heading directions θ .

Constrained evasion strategies. We next incorporated the physical constraints on θ imposed by the C-start biomechanics into

our evasion strategies. To this end, we mapped the response angle $\theta_i^{(n)}$ predicted by evasion strategy n onto the interval $[0, 100^\circ]$ using the quadratic mapping

$$\theta \rightarrow \left[1 - \left(1 - \frac{\theta_{\max}}{\pi}\right) \frac{|\theta|}{\pi}\right] \theta. \tag{12}$$

Small turns get less constrained, whereas large turns are limited to the maximum angle $\theta_{\max} = 100^\circ$ allowable by the fish biomechanics. We applied this constraint to the three most plausible strategies: distance-optimal, orthogonal, and antipodal. For each constrained strategy, we repeated the analysis presented above under precise and noisy sensing and response. Results of the K–L divergence and AIC analysis for the constrained strategies are shown in Fig. 6. Compared to the unconstrained strategies, penalizing large turns makes all three strategies fit better the experimental data across all datasets, with or without added noise, with the exception of the distance-optimal strategy for the mid-speed predator. Under precise sensing and response, the relative ranking of the constrained strategies (Fig. 6A) is similar to the original ranking (Fig. 4A), with the distinction that the orthogonal strategy at slow and mid-speed predator speeds surpasses the distance-optimal strategy and becomes the best ranking model. Under noisy sensing and response, the antipodal strategy ranks higher than the distance-optimal strategy in all but the slow predator dataset (Fig. 6B). Importantly, whether precise or noisy, the orthogonal strategy fits the experimental data better than the other two in all datasets.

Discussion

We developed a comprehensive framework for resolving evasion strategy from kinematic measurements. Our approach considers multiple hypotheses, each defined mathematically (Fig. 1), that address the role of sensorimotor noise (Fig. 3) and incorporate the effects of biomechanical constraints (Figs. 5–6). Importantly, our approach provides a rigorous methodology, rooted in strong-inference principles (27), for revealing the strategy that best fits previous kinematic measurements of zebrafish larvae. This approach eliminates bias toward a particular hypothetical strategy, as done in a previous study that favored the contralateral strategy from the dataset presently analyzed (17).

We found that the responses of zebrafish larvae to evade a predator are best characterized by the orthogonal strategy (Fig. 4). This finding challenges the notion that a prey aims either to solely confuse, or maximize its distance from, the predator with its escape (16). The kinematics of zebrafish do not exhibit the uniform distribution of escape direction θ characteristic of a pure-protean strategy (SI Appendix, Fig. S2E) (13–15). Instead, larvae exhibited correlations between θ and predator state, including angular position ϕ (SI Appendix, Fig. S5) and heading ψ (SI Appendix, Fig. S7). The distance-optimal strategy is more predictive of zebrafish kinematics but is inferior to the orthogonal strategy, based on K–L divergence and the AIC scores (Figs. 4 and 6). Therefore, zebrafish larvae do not conform to the classic dichotomy of models for prey strategy. Although the prevailing patterns favor an orthogonal strategy, variation about the predictions for this hypothesis allows for the possibility of a mixed strategy that could hinder a predator’s ability to anticipate the prey’s direction. These results are relevant to predator–prey encounters, and hence the ecology, of fish species and reflect the advantages and constraints of the prey’s neurophysiology and biomechanics.

The distance-optimal strategy requires sensing that may exceed the abilities of larval fish. This strategy requires detection of the speed of the approaching predator (Table 1), but larval fish possess poor visual acuity, compared to adult fish, due to a relatively small number of retinal cells (41). It has been demonstrated that the escape is triggered by a threshold diameter of a looming visual stimulus, which may be simulated as a circle with an expanding diameter (19). A looming stimulus alone does not offer the means to differentiate between threats that are small and fast or large and slow. Therefore, the visual system of larval fish may offer a sensory constraint on its ability to perform the distance-optimal strategy. A more sophisticated visual system could allow for additional cues to gauge the speed or size of a predator, but the processing time necessary to formulate a distance-optimal response may still pose a liability in evasion speed compared with the orthogonal strategy.

The orthogonal strategy merely requires an estimate of the predator’s heading and offers tactical benefits relative to many of the alternatives. This strategy is equivalent to the distance-optimal strategy for a high-speed approach ($U/V \ll 1$) and therefore succeeds in maximizing the prey’s distance from a fast predator at reduced sensing requirements (Table 1). It is the

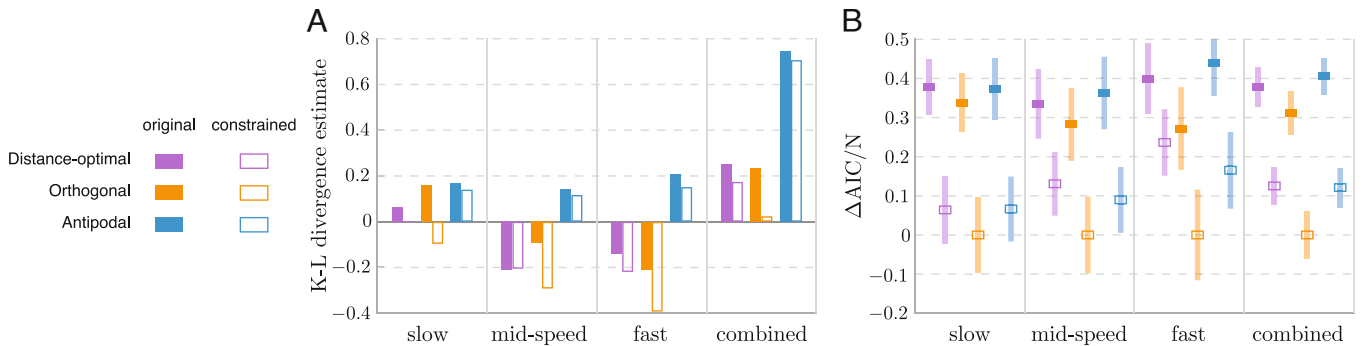


Fig. 6. Evaluation of the constrained strategies that consider the physical constraint on turning. Results are shown for the three best-performing models. (A) K–L divergence estimates of the constrained models with precise sensing and response (hollow bars), shown with the results of the original models (solid bars, from Fig. 4A). In all four datasets, the constrained models provide discernibly lower K–L divergence estimates and thus better fit to experimental data than the original models, except the distance-optimal strategy for the mid-speed predator. The orthogonal strategy improved the most after imposing the constraint, making it fit the data best in all datasets. (B) The normalized relative AIC for the constrained models with optimized noise in sensing and response, compared to results using the original models in Fig. 4B. The orthogonal strategy still provides the best fit to all datasets, marked by the lowest AIC scores, and its advantage over the second best model is more noticeable after imposing the constraint. The constrained antipodal strategy performs comparable to or even better than the constrained distance-optimal strategy.

fastest predators that likely present the greatest threat to the prey. The orthogonal strategy offers an additional tactical advantage by evading in a direction that is challenging for a fast-approaching predator to follow because, in order for the predator to execute such large turn at high speed, it needs a large turning radius, which could increase its distance from the prey even further.

The predictions of the orthogonal strategy improved in their fit to measured kinematics when we considered constraints imposed by the biomechanics of the C-start (Fig. 6). In particular, our model of a three-link fish in potential flow accurately describes the relationship between the change in fish shape and its turning motion during evasion (Fig. 5). By mapping maximum bending angle to turning angle, the model predicted an upper limit (around 100°) on achievable turning motion, consistent with the maximum angle observed in zebrafish exposed to a lateral looming stimulus (19, 23). The improvement in model predictions that included mechanics demonstrates the influence of the constraints imposed by the prey biomechanics and its interaction with the fluid environment on the evasion strategy of zebrafish larvae.

Comparing the K–L divergence and AIC values across slow, intermediate, and fast predators, the prevalence of the orthogonal strategy is clearest in the case of the fast predator (Figs. 4 and 6). This feature can be related to the fact that a weak stimulus (slow predator) is more likely to trigger an escape response via the less predictable, long-latency neural pathway (42, 43), as opposed to the fast pathway with minimal latency between perceived danger and motor response (44). An untangling of these features requires a deeper investigation of how our analytical framework relates to the neurophysiology underlying zebrafish evasion.

Our study combined tools from information theory and probabilistic methods with behavioral evasion models and physics-based models of the C-start biomechanics to develop a comprehensive analytical approach and thereby determine the evasion strategy of zebrafish larvae. Aside from the details

of the biomechanics model, nothing about our approach is specific to the study of fish. Our analysis could be applied to the myriad of studies that have measured escape responses relative to a predator's approach in a diversity of animals (5–10). This approach may therefore be applied broadly to the study of predator–prey encounters to reveal the strategic basis of this fundamental aspect of animal behavior.

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*. The relevant data and code is available at <https://github.com/ekanso/EvaluatingEvasionStrategies>. Previously published data were used for this work. We used the raw data that was previously analyzed and published in (17). We analyzed the data from scratch, so-to-speak, and we do not repeat or reuse any of the figures published in this previous work.

ACKNOWLEDGMENTS. E.K. acknowledges support from the Office of Naval Research (ONR) Grants N00014-22-1-2655, N00014-19-1-2035, N00014-17-1-2062, and N00014-14-1-0421; the National Science Foundation (NSF) Grants RAISE IOS-2034043, CBET-2100209, and INSPIRE MCB-1608744; the National Institutes of Health (NIH) Grant R01 HL 153622-01A1; and the Army Research Office (ARO) Grant W911NF-16-1-0074. This research started in summer 2018 at the Summer Graduate School on Mathematical Analysis of Behavior organized by Ann Hermundstad, Vivek Jayaraman, Eva Kanso, and L. Mahadevan. The school was jointly supported by the Mathematical Science Research Institute (MSRI) and the Howard-Hugh Medical Institute (HHMI) Janelia Research Campus and was held at Janelia. B.C., Y.M., and E.K. would like to thank Shashank Pisupati and Ann Hermundstad for helpful discussion while attending the school.

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